

Problem Statement

Describe the problem that the selected project is trying to solve.

Mental health conditions in Nigeria and across Africa remain vastly underdiagnosed due to limited access to trained mental health professionals, geographic barriers to care, and persistent stigma. Many individuals--particularly in rural and underserved urban communities--struggle to articulate their mental health needs, and existing systems lack the capacity to detect psychological distress from informal communication channels such as social media, SMS, community forums, and family conversations. This diagnostic gap means early intervention opportunities are missed, vulnerable populations cannot access timely support, and untreated mental illness escalates into crisis, loss of productivity, self-harm, and suicide. Latent Probing for Mental Health Sentiment Classification addresses this by developing a specialized language model that can identify nuanced emotional and psychological indicators from unstructured text in Nigerian and African contexts, enabling early detection, risk stratification, and referral pathways for individuals who would otherwise remain invisible to the formal healthcare system.

Who is affected by this problem?

Individuals experiencing mental health challenges in low-resource settings across Nigeria and sub-Saharan Africa; adolescents and young adults navigating identity, trauma, and social pressure without professional support; survivors of violence, conflict, and displacement; persons with chronic illnesses facing psychological comorbidities; families unable to afford formal mental health diagnosis or treatment; healthcare workers in rural clinics and community health centers lacking diagnostic tools and training; overburdened mental health professionals operating far beyond capacity; and marginalized populations--including LGBTQ+ individuals, persons with disabilities, and ethnic minorities--who face additional barriers to care due to stigma and discrimination.

Why is solving this problem important?

Without scalable early detection systems, untreated mental illness cascades into crisis: depression worsens into suicide ideation, anxiety disorders prevent economic participation, trauma compounds intergenerational harm. In Nigeria, mental health services are geographically and economically concentrated in urban centers, leaving 80% of the rural population without access. A single psychiatrist may serve over 500,000 people, however the human cost is immense. Nigeria has one

of the highest suicide rates in Africa; youth mental health crises are rising; and trauma from insecurity compounds existing mental health burdens. A scalable, AI-driven sentiment detection tool democratizes access to mental health screening, reduces the diagnostic burden on scarce professionals, enables early intervention before crisis, and creates pathways to timely support--potentially saving lives, reducing long-term disability, and enabling economic productivity and social stability across communities.